Do Brands Sound the Way They Talk?

Capstone in Data Science

Hudson Colletti

November 3, 2026

Introduction

Prior Studies and Background

Previous research on music as a form of atmospherics breaks apart into many different studies but converges to the consistent idea that music can shape how customers feel and ultimately behave. In retail and economic settings, strategic music choice has proven to differ from venue to venue. This section reviews three core studies that frame this project and introduces a gap that this project addresses. Hoffman and Turley (2002) drive the theoretical framework, Ronald Milliman's (1986) background music experiments on tempo complement theory with empirical work, and Areni and Kim (1993) provide more complex empirical research on how genre and venue can affect spending behavior. Rather than summarizing each paper independently, this review will look holistically at all three together by common themes.

Each paper defines atmospherics in a slightly nuanced manner, yet all connect to an overarching notion of influence into an individual's feelings. Given the theoretical nature of the paper, Hoffman and Turley (2002) provide the most detailed interpretation of atmospherics as the intentional control/manipulation of environmental cues within a business such as interior and exterior design, lighting, color, temperature, scent, and music. Philip Kotler (1973) was one of the first to study atmospherics, with which he argued that a store's "felt" but not always seen experience can be more important to a purchase than a product on its own. It is worth noting that in physical retail interactions consumers never experience a product apart from the environment around it. Milliman's study (1986) opens with acknowledgement of an atmosphere's importance before empirically showing the statistical significance of such.

Broader psychological research assumes that emotions and feelings are the driving forces in determining what people do and how they do it. In order to explain how subconscious responses to changes in an atmosphere work, Hoffman and Turley (2002) implemented the Stimulus-Organism-Response (S-O-R) model. Extracting from the name quite literally, a stimulus in any given environment, like music for example, acts on a person and ultimately produces an emotional and behavioral response. A consumer's reactions to environmental stimuli fall along three emotional dimensions: pleasure versus displeasure, arousal versus non-arousal, and dominance versus submissiveness (Mehrabian and Russell 1974; Hoffman and Turley 2002). Together, the emotional states of each reaction contribute to either approach or avoidance behavior. More specifically, these emotional states can guide a consumer to want to remain in an environment rather than leave, communicate with employees or other customers, and most importantly feel satisfied with their experience. Therefore, music is relevant not only in the sense of what customers buy but whether they want to be in the environment or store to begin with. The most important argument for this project from Hoffman and Turley is that atmospherics can communicate something about the store image to a consumer. Music can produce a signal as to whether a store's environment is intended to feel relaxed or energetic, social or individual, and a multitude of other feelings. While previous research does not

test whether a company's music matches its stated or intended identity, it does provide the theoretical basis for asking that question.

Through decades of research, scientists have established that in economic settings, the volume and the presence versus absence of music more generally change consumer behavior. As Milliman (1986) introduces his empirical research, the paper briefly summarizes psychology-focused surveys on the interaction of music and customer spending. The studies found that about 70 percent of consumers prefer that stores have background music inside their atmosphere. Further, without mention of what type of music was played, surveys also found that customers bought more as a result of the background music. Milliman also highlighted findings from Smith and Curnow (1966) where loudness of background music was experimented with in large supermarkets. As a result, shoppers spent significantly less time in the store while loud music (as opposed to soft music) was playing, and the rate of the spending over the loud and soft music periods was greater during the loud session. Eventually, Smith and Curnow proposed the "arousal hypothesis," where loud background music resultantly increases a person's physiological arousal and pace of activity/movement. This notion of arousal draws quite close to the emotional dimension described in the S-O-R model previously mentioned and thus may prove informative in this project.

Drawing from the findings of previous research, Milliman conducted a study in 1982 to experiment with varied levels of tempo on consumer behavior. In this initial study, music tempo (slow versus fast) was chosen as the independent variable for the environment of a medium-sized supermarket. In his 1986 paper, Milliman explained that slow-tempo background music produced a significantly slower pace of in-store traffic flow and a significantly greater sales volume. Looking to show statistically significant effects in a second study, Milliman redesigned the study to observe a medium-sized restaurant in Dallas/Fort Worth, Texas. This specific restaurant was priced above average, appealed to middle-aged people in the upper-middle income bracket, typically held 30-60 minute wait times on Fridays and Saturdays, and groups averaged 4.25 people, 77% of whom drank alcoholic beverages (Milliman 1986).

Before stepping into the restaurant, Milliman conducted a survey to define the tempo variable values. Respondents perceived slow tempo music as having 72 or fewer beats per minute (bpm) and fast tempo music as having 92 or more bpm. Additionally, to remove biases that individuals may have for popular songs or the gender of a singer, Milliman played only instrumental music. In most restaurant outcomes tested, Milliman observed a statistically significant difference between the two treatment groups. Similar to his previous experiment, the time spent dining on average was higher (56 minutes) with slow tempo music playing compared to fast tempo music (45 minutes) at a significance level of $p < 0.01$. Likely correlated, the wait before being seated followed the same trend with slow tempo music averaging a wait time of 47 minutes and fast tempo averaging a quicker 34 minutes ($p < 0.01$). Additionally, the amount spent on alcohol with slow tempo music playing was \$8.85 more than the fast tempo treatment group on average ($p < 0.01$). Finally, the gross margin per bill between the groups was also observed at a significant difference. The average gross margin per statement was \$55.82 under slow-music treatment versus \$48.62 under the fast-music treatment ($p < 0.05$) (Milliman 1986).

Despite these observed differences, the amount spent on food and the number of parties that left while waiting for a table did not differ significantly between the treatments. Groups typically stayed longer and ordered an average of 3.04 more drinks. Milliman ultimately interpreted these observations to believe that slow tempo music created a more relaxing environment, or an approach response in terms of S-O-R. Through the differences in the two treatment groups, noticeable tradeoffs for store owners exist: higher margins and revenue from alcohol came with longer wait times. Thus, the tempo of background music is an option for business owners to strategically achieve a business goal. If a company values customer experience/relaxation over simplicity and shorter wait times, then there are obvious links to what tempo background music that business should play in its physical locations.

Another variable worth studying for the scope of this project is genre. Work from Areni and Kim (1993) introduces an idea that plays a crucial role in this project, predicting that music may be most effective atmospherically when it aligns with the context in which it is played. Put simply, a spa plays relaxing music rather than an overstimulating song you may hear at a rave. With a similar perspective, Areni and Kim took their research to a wine store where they experimented with both classical and Top-Forty music. The authors believed classical music was especially appropriate for a wine store because both wine and classical music are commonly associated with sophistication, prestige, foreignness, and art/culture. As a result, the classical music could "reinforce" the atmosphere consumers associate with purchasing wine while Top-Forty music might be less fitting.

After randomly assigning classical and Top-Forty music across nights within the wine cellar of a restaurant in the southeastern United States, researchers observed customers while pretending to act as staff performing inventory work in the cellar. Genre ultimately had no significant effect on the total number of items customers picked up or purchased and did not change the amount of time they spent inside the cellar. However, when analyzing the purchases that did occur, the average sale under classical music was \$7.43 compared to \$2.18 under Top-Forty music, meaning there was a difference in the amount of money spent ($p < 0.02$). After repetition and controlling for customer demographic, the difference remained statistically significant. Thus, classical music did not lead to customers purchasing higher volume but rather more expensive items. Areni and Kim suggest that customers may have experienced the classical music as attributing to an upscale environment and they purchased accordingly to that atmosphere. This change is likely most evident in customers with limited wine knowledge as a more experienced customer would be less wavering in the amount spent on wine dependent on an atmosphere. The study only observes the two included genres and the singular wine cellar but still shows that music can shape the appearance of a store and the spending tendencies of a customer (Areni and Kim 1993).

Together, these studies demonstrate that music shapes/adjusts consumer behavior through variables like loudness, tempo, and genre. Further, the extent to which each effect is observable also depends on how well the music aligns with the setting it is played within. However in each paper, the researchers are only examining one venue/brand and the music that is played is selected by the researchers, not the venue itself. When considering the alignment of music

with the setting it is played within, analyzing the music that companies specifically choose to play may address more complexities within this field. This project bridges gaps within the culmination of these studies through comparing 30 venues at once and looking at both the retailer's identity and the music connected to their brand. The goal is to observe and measure the alignment between a retailer's stated identity and the musical atmosphere rather than the causal effect of music on spending because previous work has substantively studied cause and effect within the realm of this topic.

Additionally, Hoffman and Turley's pleasure and arousal dimensions roughly correspond to the track-level metrics like valence and energy that Chosic provides for the music data utilized in this project. A recent validation study strengthens this correspondence by finding a strong association between Spotify's energy measure and listeners' arousal ratings. Furthermore, the study also found that Spotify's valence measure was moderately associated with listeners' mood ratings (Vidas et al. 2026). Chosic's platform is an app that utilized the Spotify API before usage rules were updated, so the app acts as an equivalent to the API. Thus, the results from the Spotify validation study should transfer directly to Chosic.

Data Sourcing and Company Selection Pipeline

Breaking Down the Mood of Brand Language and Music

Visualizations & Dashboard

Modeling Approach

Results

Ethical Considerations

Conclusion and Limitations

References

Areni, Charles S., and David Kim. 1993. "The Influence of Background Music on Shopping Behavior: Classical Versus Top-Forty Music in a Wine Store." *Advances in Consumer Research* 20: 336–40.

Hoffman, K. Douglas, and L. W. Turley. 2002. "Atmospherics, Service Encounters and Consumer Decision Making: An Integrative Perspective." *Journal of Marketing Theory and Practice* 10 (3): 33–47.

Kotler, Philip. 1973. "Atmospherics as a Marketing Tool." *Journal of Retailing* 49 (4): 48–64.

Mehrabian, Albert, and James A. Russell. 1974. *An Approach to Environmental Psychology.* MIT Press.

Milliman, Ronald E. 1986. "The Influence of Background Music on the Behavior of Restaurant Patrons." *Journal of Consumer Research* 13 (2): 286–89.

Smith, Patricia Cain, and Ross Curnow. 1966. "'Arousal Hypothesis' and the Effects of Music on Purchasing Behavior." *Journal of Applied Psychology* 50 (3): 255–56.

Vidas, Dianna, Lewis Nitschinsk, Margaret S. Osborne, and Nikki S. Rickard. 2026. "Validating Spotify's 'Valence,' 'Energy,' and 'Danceability' Audio Features for Music Psychology Research." *Music Perception: An Interdisciplinary Journal* 44 (1): 40–53. https://doi.org/10.1525/mp.2026.2463161.